Lecture 38: Deep Learning (Part-3) — ANN Regression aur Multi-Class Classification (Hinglish Master Guide)

Deep Learning Curriculum ke **Lecture 38** mein aapka swagat hai! Is lecture mein humne **PyTorch** ka upyog karke do comprehensive deep learning industrial projects scratch se implement kiye hain:

1. **Continuous Regression Project**: Combined Cycle Power Plant ke sensor data se electrical energy output (MW) predict karna.
2. **Multi-Class Classification Project**: Khajur (Date Fruit) ke 34 physical features se uski 7 varieties ko classify karna.

---

1. Master Curriculum aur Submodules Index

Submodule	Topic	Key Concepts Covered	Hand-Drawn Educational Diagram
Notes_38.01	**ANN Regression Formulation**	Power plant problem, continuous output, architecture ($4 \to 64 \to 32 \to 1$), parameter count derivation	`ann_regression_architecture.png`
Notes_38.02	**Data Loading & Preprocessing**	`powerplant_data.csv`, `StandardScaler`, data leakage se bachav, float32 conversion	`regression_preprocessing_pipeline.png`
Notes_38.03	**TensorDataset & DataLoader**	Tensors bundling, mini-batching ($B = 32$), shuffling mechanics, memory optimization	`tensordataset_dataloader_pipeline.png`
Notes_38.04	**PyTorch Model Construction**	Subclassing `nn.Module`, `nn.Sequential`, `nn.Linear`, `nn.ReLU`, linear output mapping	`pytorch_nn_module_anatomy.png`
Notes_38.05	**Training & Validation Loop**	Optimization ke 4 steps: `zero_grad()`, `forward()`, `MSELoss()`, `backward()`, `step()`	`pytorch_training_validation_loop.png`
Notes_38.06	**Model Checkpointing**	Overfitting mitigation, `model.state_dict()`,	`model_checkpointing_workflow.png`

Submodule	Topic	Key Concepts Covered	Hand-Drawn Educational Diagram
	& `state_dict`	dictionary saving, best validation loss selection	
Notes_38.07	**Regression Model Evaluation**	`model.eval()`, `torch.no_grad()`, MSE (17.81), RMSE (4.22 MW), R^2 Score (0.937)	`regression_evaluation_metrics.png`
Notes_38.08	**ANN for Classification: Formulation**	Date fruit dataset (898 samples, 34 features, 7 classes), output layer sizing ($C = 7$)	`ann_classification_architecture.png`
Notes_38.09	**Multi-Class Preprocessing Pipeline**	`LabelEncoder` ($0 \dots 6$), feature scaling, `torch.long` target dtypes, `shuffle=True`	`classification_preprocessing_pipeline.png`
Notes_38.10	**Classification Model & Evaluation**	Architecture ($34 \to 64 \to 64 \to 7$), why NO Softmax in model, `CrossEntropyLoss`, `torch.max`, 93.8% accuracy	`cross_entropy_accuracy_evaluation.png`

---

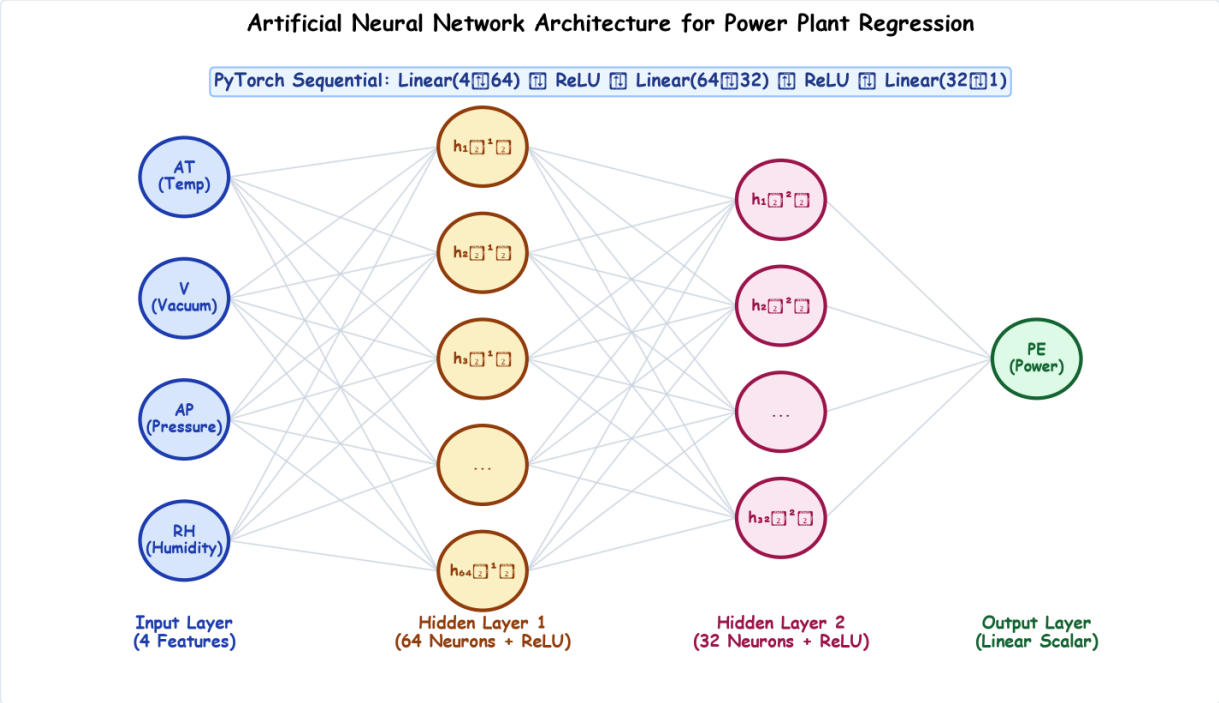
2. Regression vs Classification Architecture Comparison

```
+-----------------------------------------------------------------------------------------------
|                                    ANN ARCHITECTURE COMPARISON
+--------------------------------------------------+--------------------------------------------
| REGRESSION ANN (Power Plant)                     | MULTI-CLASS CLASSIFICATION ANN (Date Frui
+--------------------------------------------------+--------------------------------------------
| Target: Hourly Electrical Energy Output (PE)     | Target: 7 Date Fruit Varieties (0 to 6)
| Input Dimension: 4 Ambient Features              | Input Dimension: 34 Morphological Feature
| Hidden Layers: [64 (ReLU), 32 (ReLU)]            | Hidden Layers: [64 (ReLU), 64 (ReLU)]
| Output Layer: 1 Neuron (Linear Identity)         | Output Layer: 7 Neurons (Unnormalized Log
| Loss Function: Mean Squared Error (nn.MSELoss)   | Loss Function: Cross Entropy (nn.CrossEnt
| Target Tensor Dtype: torch.float32               | Target Tensor Dtype: torch.long
| Total Trainable Parameters: 2,433                | Total Trainable Parameters: 6,855
| Evaluation Metric: R² Score (0.937), RMSE (4.22 MW)| Evaluation Metric: Accuracy (93.8%)
+--------------------------------------------------+--------------------------------------------
```

---

3. Hand-Drawn Educational Diagram

Neeche diya gaya diagram ANN Regression model ke complete architecture aur feature flow ko depict karta hai:



---

4. Practice Suite aur Resources

- **Practice Problems (English)** | **Practice Problems (Hinglish)**
- **Complete Solutions (English)** | **Complete Solutions (Hinglish)**
- **Central Code Repository**: `lecture_38/Lecture_Code/` jismein `ANN_Regression.ipynb`, `ANN_Classification.ipynb`, `powerplant_data.csv`, aur `DateFruit_Dataset.csv` shamil hain.